

Seq66 v. 0.99.27+
MIDI Learn with the Akai MPK Mini Play

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Akai MPK Mini Play

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1 Introduction

This document is a tutorial about using the MIDI Learn feature of *Seq66*. It supplements the "MIDI Learn" section of the user manual with a real use-case. We recommend reading through that manual section first.

The following projects support *Seq66* and its documentation (see the **Help** menu):

- <https://github.com/ahlstromcj/seq66.git>
- <https://ahlstromcj.github.io/>
- <https://ahlstromcj.github.io/docs/seq66/akai-mpk-mini-play-learn.pdf>

1.1 Background

The Akai MPK Mini Play Mk3 is an interesting device. It has a full-fledged MIDI engine, but uses an ARM processor that constrains what can be done with it. The following article explains how it works.

<http://sandsoftwaresound.net/akai-mpk-mini-play-midi-implementation/>

When attached via USB, channel 10 is used for percussion, all other channels respond to notes, but all program change events are redirection to channel 1 (which drives one nuts until one learns about it).

This tutorial describes using the 16 pads (8 "A" pads and 8 "B" pads) to mute/unmute *Seq66* pattern slots. It assumes that drum set 001 "Std. Set" is selected, to make sure the pads always

emit the same set of events. (We will look into the possibility of creating a macro to select that drumset.)

For *Seq66* and MIDI, what matters is that the MIDI controller portion of the MPK Mini emits events only on channels 1 and 10. Also, the ARM mini-controller filters out all SysEx other than its own.

This tutorial assumes that Linux ALSA, JACK, and Windows MIDI have been set up properly for use with *Seq66*.

1.2 Setup

Stop *Seq66*. Plug in the MPK Mini. Run *Seq66* and verify that it is active in the **Edit / Preferences / MIDI Clock** and **MIDI Input** tabs and the The setup involves the following steps:

1. Create a 'ctrl' file.
2. Specify it and activate it in the 'rc' file.
3. Set the MPK Mini as an input controller.

1.2.1 Setup / 'ctrl' File

Copy the `data/linux/qseq66.ctrl` to a file named after the device, such as `akai-mpk-mini-play.ctrl`. Copy this file to "\$home", which is usually `~/.config/seq66`.

Read up on the '*ctrl*' File section of the user manual to get familiar with the control sections of this file. The sections relevant to MIDI Learn are

```
[comments]
[loop-control]
[mute-group-control]
[automation-control]
```

Only the comments section needs manual editing (for an explanation of the intended usage of the file). The rest are covered by the **Edit / Preferences / Session** tab and the **Learn** dialog, though the 'ctrl' can still be manually edited, if desired.

At this point, the 'ctrl' file is in place, but not yet ready. It needs to be specified in the 'rc' file.

1.2.2 Setup / 'rc' File

In the **Edit / Preferences / Session** tab, set up the new 'ctrl' file as shown here.

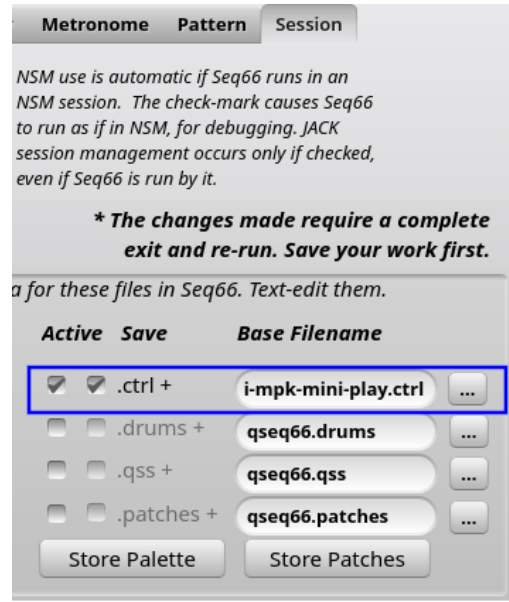


Figure 1: Set New 'ctrl' File

Note that it is marked as active and to be saved. Exit *Seq66*.

In the 'rc' file `~/.config/seq66/qseq66.rc`, one should see the following items (adapted to one's personal setup):

```
[midi-control-file]
active = true
name = "akai-mpk-mini-play.ctrl"
```

Now restart *Seq66*. The changes made in the next section will now be able to be written to the 'ctrl' file.

1.2.3 Setup / I/O Ports

Refer to the following figures.

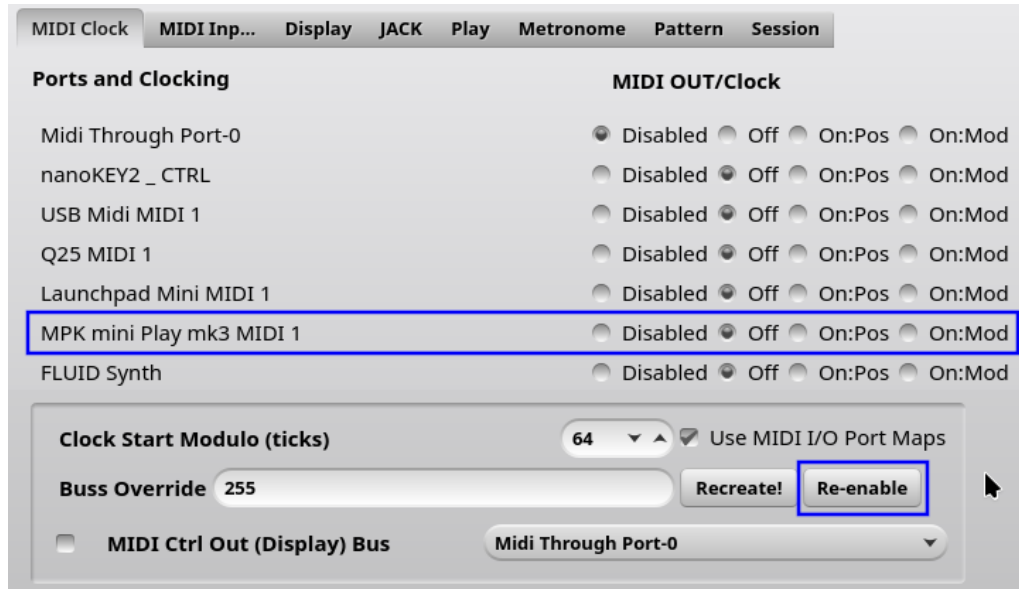


Figure 2: Set MPK Mini Output Status

Some ports might be disabled, even though present, because they were missing. If it is known they are plugged in click, click the **Re-enable** button. This also enables the input port entries in the **MIDI Input** tab. Set the MPK Mini controller as desired (usually "Off").

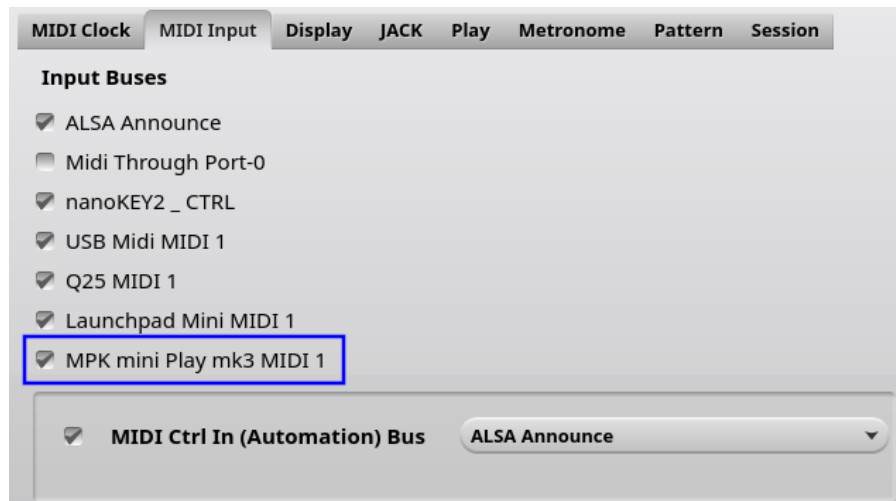


Figure 3: Set MPK Mini Input Status

Make sure the MPK Mini is enabled. Do not set **MIDI Ctrl In** at this time. Restart *Seq66* and verify that the MIDI clock and input tabs show the correct settings.

In the 'rc' file `~/.config/seq66/qseq66.rc`, one should see the following items (adapted to one's personal setup):

```
[midi-input]
...
```

```

6 1 "[6] 48:0 MPK mini Play mk3 MIDI 1"

[midi-input-map]
1      # map is active
. . .
6 1 "MPK mini Play mk3 MIDI 1"

[midi-clock]
. . .
5 0 "[5] 48:0 MPK mini Play mk3 MIDI 1"

[midi-clock-map]
1      # map is active
5 0 "MPK mini Play mk3 MIDI 1"

```

If not, this is an issue and should be reported. In the meantime, one can try to edit the 'rc' file.

1.2.4 Setup / I/O and Automation Control

Once all the ports are configured, one can then select the MPK Mini as the control/automation buss.



Figure 4: Set MPK Mini Input as the MIDI Controller

In the 'ctrl' file `~/.config/seq66/akai-mpk-mini-play.ctrl`, one should see the following items (adapted to one's personal setup):

```

[midi-control-settings]
control-buss = "MPK mini Play mk3 MIDI 1"
midi-enabled = true

[midi-control-out-settings]
output-buss = "Midi Through Port-0"
midi-enabled = false

```

1.3 MIDI Learning

Now we are ready to make the percussion pads control 16 pattern slots. Start *Seq66* and then select **Learn / Loops** from the main menu. The learn manager that comes up should look blank, like the following. If not, the wrong 'ctrl' file has been configured in the 'rc' file. Try again with section [1.2.2 "Setup / 'rc' File"](#), page [3](#).

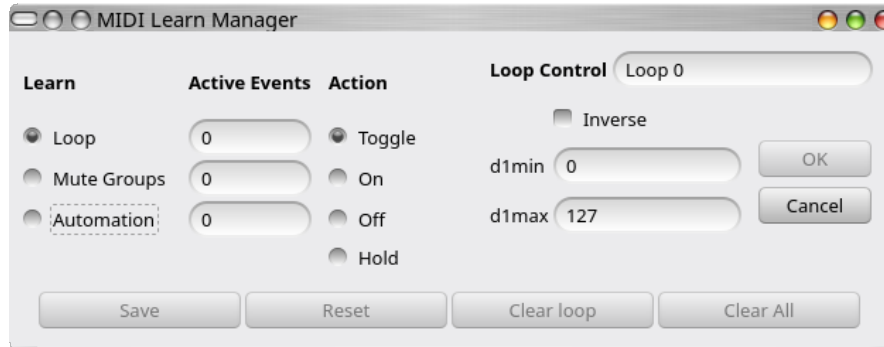


Figure 5: Blank/Initial Learn Manager

Without doing anything else, go to the Mini Play and strike each pad in group "A" once, in order. Then select group "B" and strike each pad again, in order.

After these actions, the **Loop** count should be 16, and the **Save** button should be enabled. Click it. This saves the file specified for 'ctrl' in **Edit / Preferences / Session** *immediately*.

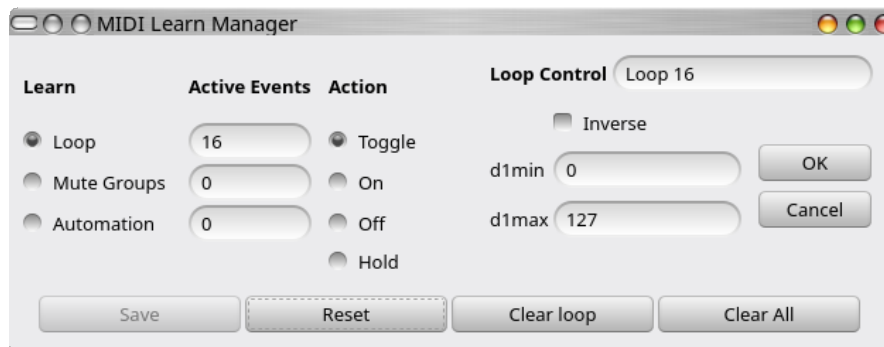


Figure 6: Learn Manager After Save

One can look at the `akai-mpk-mini-play.ctrl` file and verify in the `[loop-control]` section that the first 16 lines in the "toggle" section show active events such as:

```
0 "1"  [ 0 0x99  40  0 127 ] [ 0 0x00  0  0  0 ] ...
1 "q"  [ 0 0x99  41  0 127 ] [ 0 0x00  0  0  0 ] ...
2 "a"  [ 0 0x99  42  0 127 ] [ 0 0x00  0  0  0 ] ...
. . .
```

where the first number is the slot number, the character in quotes is the hot-key, and the first set of

numbers are the "inverse" flag, 0x99 is a note-on for channel 10, next is the note value, and 0 and 127 is the acceptable range of note velocities.